Course Notes for SOC 527 (Social Networks)

This is a course handout for the lecture on stochastic actor-oriented models (SAOMs) for SOC 527: Social Networks at the University of Arizona. The course handout provides a brief technical introduction for stochastic actor-oriented models and a (rather incomplete) reference list that details the methodological development of the SAOM. Importantly, this course handout should be read alongside the lecture slide deck and the course R code script. In this way, (1) the slide deck provides a general understanding and example empirical applications of the SAOM, (2) the R code file explains how to estimate an SAOM given empirical data using the RSiena R package, and (3) the course handout provides a more technical introduction for the SAOM that goes beyond the lecture slide deck.

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What is the stochastic actor-oriented model?

The **stochastic actor-oriented model (SAOM)** is an inferential continuous-time markov model that allows us to understand how a network evolves (changes) across waves of data collection. Snijders (1996, 150) developed the **SAOM** to create a "statistical model that is itself a direct expression of ... sociological theory" to study the evolution of networks. To accomplish this, Snijders (1996) combined methodological individualism with continuous-time markov models. In the methodological individualism theoretical framework, agents are assumed to be purposive actors who choose social actions that maximize their 'utility' based upon a set of constraints. In the next section, we outline continuous-time markov models for studying the evolution of social networks across time and then outline the stochastic actor-oriented model. This document is only a brief introduction to these modeling frameworks, for more information on the models proposed here, please see Table 1.

Continuous-time markov models for the evolution of social networks

Continuous-time markov models for social network dynamics allow us to study the dynamic

processes associated with the creation and cessation of social ties over time and were developed in the foundational work of Holland and Leinhardt (1977), Wasserman (1980), and Leenders (1995). Following Holland and Leinhardt (1977), let $X(t)$ be a continuous-time stochastic process representing a $g \times g$ binary sociomatrix where social ties among g actors evolve over time. The stochastic process at time t can be visualized as:

$$X(t) = \begin{bmatrix} 0 & X_{12}(t) & X_{13}(t) & \cdots & X_{1g}(t) \\ X_{21}(t) & 0 & X_{23}(t) & \cdots & X_{2g}(t) \\ X_{31}(t) & X_{32}(t) & 0 & \cdots & X_{3g}(t) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ X_{g1}(t) & X_{g2}(t) & X_{g3}(t) & \cdots & 0 \end{bmatrix} \tag{1}$$

where the $(i, j)^{\text{th}}$ entry is:

$$X_{ij}(t) = \begin{cases} 1 & \text{if } i \to j \text{ at time } t \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

In addition, self-loops are not allowed (i.e., the diagonal values of $X(t)$ are set to 0) and the entries are directed ties. Moreover, it is assumed that $t \in \mathbb{R}$ is a continuous time parameter with $t \geq 0$ (Snijders, 2001; Leenders, 1995). This assumption implies that changes in the network structure can occur at any point in continuous time, rather than at discrete time points, and that the timing between network changes is stochastic (random). Moreover, let $X(t) = x$ be the realized sociomatrix at time t. In this document, lower case variables, x, denote realizations of a random process/variable, X. Following Holland and Leinhardt (1977), S is the state space which contains all possible $g \times g$ binary sociomatrices with $x, y, z, k, \cdots$ being possible sociomatrices in S. In other words, S is the set of possible values the stochastic process, $X(t)$, can take across $\mathbb{R}$. The size (cardinality) of S is $2^{g(g-1)}$. For instance, the size of the state space for a process with 5 actors is $2^{20} = 1,048,576$!

To understand the evolution of the stochastic process $X(t)$ over time (i.e., why the realized

values from $X(t)$ takes certain values from S across time), Holland and Leinhardt (1977) postulated two assumptions: (1) $X(t)$ is a Markov process and (2) the conditional independence of subsequent tie changes.

Assumption 1. $X(t)$ is a Markov process, such that, the distribution of future network state $X(t+h)$ in a small time interval h is conditional upon the current network state x. More formally, the transition probability for a small interval of time h is (Wasserman, 1980):

$$\mathrm{P}_{x\to y}(t,h) = \mathrm{P}\{X(t+h) = y | X(t) = x\} \tag{3}$$

where $\mathrm{P}_{x\to y}(t,h)$ represents the transition probability of moving from the current realized network x to the next network state y across the small time interval h. This assumption implies that the probability of changing from one network state, x, to the next, y, is dependent on the current network configuration. As Holland and Leinhardt (1977, 8) noted with this assumption: "we wish to reflect the idea that the structure of the network at time t can influence the probability of subsequent changes in its structure." In addition, the assumption asserts that the conditional distribution of future network states is not influenced by past structure of relations beyond the current configuration, x.

Assumption 2. Given the current network state, x, at time t, subsequent changes in the edge (ties) are statistically independent of each other after a small amount of time. More formally, the conditional independence of subsequent tie changes states (Holland and Leinhardt, 1977):

$$\begin{gathered} P_{x\to y}(t,h) = \prod_{i,j} P\{X_{ij}(t+h) = y_{ij} | X(t) = x\} + o(h) \\ \text{as } \; h \to 0 \end{gathered} \tag{4}$$

where $o(h)$ is a function that goes to zero faster than h, such that $o(h)/h \to 0$ as h goes to 0. The second assumption asserts that, given the current network state x at time t, the probability of going from the current network state x to the possible state y in a small time interval h is equal

to the product of conditional tie changes (Wasserman, 1980). In addition, this assumption implies that the probability of more than one tie changing simultaneously within the interval is of order $o(h)$, which is negligible in comparison to one tie changing as the time interval $h \to 0$ (Holland and Leinhardt, 1977). As Leenders (1995, 4) notes, "[this] assumption states that for very small intervals of time the changes in the digraph are statistically independent, so the probability of any two arcs changing simultaneously is much smaller than the probability of a single change. One way to think about this is to consider infinitesimally small time intervals that practically discretize time. When a change occurs it occurs so quickly that no other change can occur at the same time and no other change is influenced by it during that extremely small time interval."

Given these two assumptions, the change intensities, λ, for the process, $X(t)$, over a small time interval h is:

$$\underbrace{\mathrm{P}\{X_{ij}(t+h) = 1 - x_{ij} | X(t) = x\}}_{\text{probability that the } i \to j \text{ tie changes}} = h\lambda_{ij}(x,t) + o(h)$$

where $\lambda_{ij}(x,t)$ is characterized as a function of the current network state x and actors i and j. For example, the Wasserman (1980) **reciprocity model** is:

$$\lambda_{ij}(x,t) = \underbrace{(1-x_{ij})(\lambda_0 + \mu_0 x_{ji})}_{\text{probability of creating the } i \to j \text{ tie}} + \underbrace{x_{ij}(\lambda_1 + \mu_1 x_{ji})}_{\text{probability of dropping the } i \to j \text{ tie}}$$

where the probability of an $i \to j$ tie being created (dropped) in the small interval $(t, t+h)$ depends on the $j \to i$ edge at time t, μ_0 is the 'importance' of reciprocity for creating a tie (parameter), and μ_1 is the 'importance' of reciprocity for dropping a tie (parameter). The reciprocity model can be formalized by examining the dyadic infinitesimal generator matrix Q_{xy}, which is:

$$Q_{xy} = \begin{array}{c|cccc} & (0,0)_y & (1,0)_y & (0,1)_y & (1,1)_y \\ \hline (0,0)_x & -2\lambda_0 & \lambda_0 & \lambda_0 & \\ (1,0)_x & \lambda_1 & -(\lambda_1+\lambda_0+\mu_0) & 0 & \lambda_0+\mu_0 \\ (0,1)_x & \lambda_1 & 0 & -(\lambda_1+\lambda_0+\mu_0) & \lambda_0+\mu_0 \\ (1,1)_x & 0 & \lambda_1+\mu_1 & \lambda_1+\mu_1 & -2(\lambda_1+\mu_1) \end{array}$$

where $(i,j)_x$ is the realized $i \rightarrow j$ tie in the x network and $(i,j)_y$ is the potential $i \rightarrow j$ tie in the y network configuration. The probability matrix for the transition of the stochastic process, $X(t)$ over the small interval h is the matrix exponential, $\exp[Q(t)_{xy}]$. Readers interested in more information on the transition matrix are referred to Wasserman (1980). In summary, **continuous-time markov models** are dynamic models that allow us to study the processes associated with the creation and cessation of social ties over time. In this modeling framework, realized social networks, x, are outcomes of a continuous-time stochastic process $X(t)$. We assume that $X(t)$ is a Markov process with subsequent changes in dyadic ties being conditionally independent. Moreover, the probability, $\lambda_{xy}(x,t)$, of going from one state x to another y in some small time interval $(t+h)$ is a function of the current network state, x.

Actor-oriented framework

Recall that methodological individualism assumes that agents are considered purposive actors who choose social actions that maximize their 'utility' based upon a set of constraints. In the **stochastic actor-oriented modeling** framework, it is assumed that actors (also known as agents; e.g., adolescents in high schools) are "individually optimizing his or her own utility," where the 'utility' is based upon a set of network and exogenous constraints and the evolution of the network between waves "is the result of the actions of the individual actors in the network" (Snijders, 1996, 152). The term **actor-oriented** comes from the idea that actors within the network control their outgoing ties, that is, who they send a tie to, and that "changes in [their] ties are made ... on the basis of their and others' attributes, their position in the network, and their perceptions about the rest of the network" (Snijders, Van De Bunt and Steglich, 2010, 46). From this assumption, the SAOM assumes that actors have full information about the current state of the network (the set of relations amongst actors within the network) and the other actors' attributes. In summary, the SAOM assumes that the changes in the network structure between data collection waves are the result of an underlying latent process whereby changes are the "result of the actions of the individual actors in the network" (Snijders, 1996, 152).

Stochastic actor-oriented model

The SAOM is a two-step **change (choice) determination continuous-time markov model** where changes to the network state occur through a sequence of 'microsteps'. In the first step, actors are randomly chosen to get the opportunity to change a tie. The waiting time until an actor i is chosen to make a change to their outgoing ties is exponentially distributed with rate parameter $\lambda_i(x,z)$ (Snijders, 1996, 2001). The rate parameter, $\lambda_i(x,z)$, can be a function of the current network state x or a node-level attribute z. For example, Leal et al. (2026) test if the rate parameter is dependent upon adolescent depression, that is, are more depressed adolescents more likely to change their outgoing ties. Commonly, we assume that the rate parameter λ_i is homogeneous for all actors. The probability that an actor i is selected to make a change at the current microstep is ρ_i (Steglich and Snijders, 2022; Duxbury, 2023):

$$\rho_i = \frac{\lambda_i(x,z)}{\sum_j \lambda_j(x,z)} \tag{5}$$

which is the rate that the actor i is selected divided by the total sum of actor rates (including i). If we assume homogeneity of the rate parameter for all g actors (Snijders, 2001; Duxbury, 2023), the probability that actor i is chosen to evaluate their outgoing ties is:

$$\rho_i = \frac{1}{g} \tag{6}$$

In the second step, the chosen actor i evaluates their current outgoing ties and makes a choice about changing or not changing their ties based upon maximizing the **objective function**. More formally, the objective function is:

$$f_i(\beta, x(i \rightsquigarrow j)) \tag{7}$$

with $x(i \rightsquigarrow j)$ denoting the toggling of the x_{ij} tie from the x sociomatrix. If $x_{ij} = 1$, then $x_{ij}(i \rightsquigarrow j) = 0$ (the dropping of the tie). In contrast, if $x_{ij} = 0$, then $x_{ij}(i \rightsquigarrow j) = 1$ (the creation of the tie). As Snijders (2001, 366) notes the objective function, f_i, "indicates the degree of satisfaction

for actor i inherent in the relational situation represented by" $x(i \rightsquigarrow j)$. The objective function is a linear combination of parameters β and effects $s_{ik}(x(i \rightsquigarrow j))$ (Snijders, 2001; Duxbury, 2023). The objective function can be written as:

$$f_i(\beta, x(i \rightsquigarrow j)) = \sum_k \beta_k s_{ik}(x(i \rightsquigarrow j)) \tag{8}$$

where β_k represents the 'importance' of the network statistic s_k for the objective function and $s_{ik}(x(i \rightsquigarrow j))$ are effects that represent a network statistic based upon the potential network configuration, $x(i \rightsquigarrow j)$ (Snijders, 2001). As an example, let $x^0 = x(i \rightsquigarrow j)$, then the effects are formulated as:

- Reciprocity effect: $s(x^0) = \sum_j x^0_{ij} x^0_{ji}$
- Indegree popularity effect: $s(x^0) = \sum_j x^0_{ij} \sum_k x^0_{kj}$
  - $\sum_k x^0_{kj}$ is the indegree of the actor j who i is related to.
- Outdegree activity effect: $s(x^0) = \sum_j x^0_{ij} \sum_k x^0_{jk}$
  - $\sum_k x^0_{jk}$ is the outdegree of the actor j who i is related to.
- Transitivity effect: $s(x^0) = \sum_j \sum_h x^0_{ij} x^0_{ih} x^0_{jh}$
- Actor outdegree (ego) effect: $s(x^0) = y_i \sum_j x^0_{ij}$, where y_i is the covariate value of y for i.
- Actor indgree (alter) effect: $s(x^0) = \sum_j x^0_{ij} y_j$, where y_j is the covariate value of y for alter j.

For an effect s_{ik}, if $\beta_k > 0$, then an actor i has an increased preference for network configurations with more relations captured by the effect. For example, let s_{ik} denote reciprocity and let $\beta_k > 0$, then network configurations with more reciprocal relations are more 'preferable' to actor i.

From the definitions in equations 7 and 8, Snijders (1996) proposed that the probability that the $i \rightarrow j$ tie changes during the current microstep is:

$$p_{ij}(\beta, x) = \frac{\exp(f_i(\beta, x(i \rightsquigarrow j)))}{\sum_k \exp(f_i(\beta, x(i \rightsquigarrow k)))} \tag{9}$$

where the denominator is the set of all possible tie changes, including the network state for which the actor i makes no changes to their outgoing ties, $x(i \rightsquigarrow j) = x$. From equation 9, actor i chooses the network configuration that maximizes their objective function in comparison to all other tie changes. The above formulation is equivalent to the McFadden utility choice model in economics (Snijders, 2001). From equation 9, β_k are in log-probability ratios, that is, β_k represents the "contributions to log-probabilities of increasing the dependent variable by 1 unit when the effect is increased by 1 unit" (Ripley et al., 2026, 218). We can show this by comparing the probability that the chosen actor i chooses to a change their $i \rightarrow k$ tie, in comparison to the $i \rightarrow j$ edge. Formally,

$$\ln\left(\frac{p_{ik}(\beta_k, x)}{p_{ij}(\beta_k, x)}\right) = \ln\left(\frac{\frac{\exp(f_i(\beta, x(i \rightsquigarrow k)))}{\sum_z \exp(f_i(\beta, x(i \rightsquigarrow z)))}}{\frac{\exp(f_i(\beta, x(i \rightsquigarrow j)))}{\sum_z \exp(f_i(\beta, x(i \rightsquigarrow z)))}}\right) = \beta_k\{s_{ik}(x(i \rightsquigarrow k)) - s_{ik}(x(i \rightsquigarrow j))\} \tag{10}$$

In summary, how does the network state evolve? Based on the above formulations, changes to the network structure occur through a series of many sequential microsteps $(x^0, x^1, x^2, \cdots, x^N)$. At microstep k, an actor i is randomly chosen to make a change to their outgoing ties based upon the rate parameter $\lambda_i(x^{k-1}, z_i)$, with x^{k-1} representing the current network configuration. Given that actor i is randomly selected to evaluate their outgoing ties, the selected actor makes a change to their outgoing ties that maximizes their objective function, f_i, where the objective function is based upon a set of endogenous (network-related) and exogenous actor-level constraints. During the microstep k, the probability of a specific $i \rightarrow j$ tie changing during a microstep is, $\rho_i \cdot p_{ij}$. Following the evaluation of their outgoing ties, the network configuration is updated and the next actor is chosen based upon the distribution of rate parameters.

Method of moments estimation algorithm

Snijders (1996) and Snijders (2001) proposed to use Method of Moments estimation, in comparison to Maximum Likelihood Estimation, to estimate the parameters of an SAOM. Following Snijders (1996), let s be a $k \times 1$ vector of statistics from the observed network in wave m (for two

waves, $m = 2$) and $E_\beta(S)$ be a $k \times 1$ vector of statistics from the simulated network in wave m based upon the fitted model. We find the β vector that solves the following moment equation:

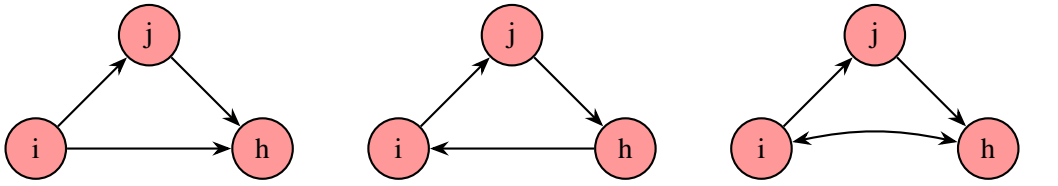
$$E_\beta(S) = s \tag{11}$$

The SAOM find the set of parameter estimates, $\hat{\beta}$, that solves the moment equation via a three-phase computer simulation based upon the Robbins–Monro approximation algorithm (Snijders, 1996). For sake of brevity, conditional upon the first network (i.e., the realized empirical network in wave 1), the SAOM simulates a Monte Carlo Markov Chain of sequential microsteps based upon the above section and updates the parameter estimates based upon the distance between the observed statistics s and the simulated statistics $E_\beta(S)$. In the first phase of the simulation algorithm, the estimation routine finds the initial set of searching parameter values. In Phase 2, based upon the first network, the SAOM simulates many, many Monte Carlo Markov Chains of microsteps to find the estimates that solve equation 11. In Phase 3, the estimation algorithm simulates an additional N Monte Carlo Markov Chains of microsteps to approximate the covariance matrix for β. More specifically, based upon those parameter values found in Phase 2, the algorithm generates an updated parameter value and uses the distribution of generated values to compute the covariance matrix for β. Readers more interested in the technical details of the simulation algorithm are encouraged to see Snijders (1996), Snijders (2001), and Ripley et al. (2026) for more information on the algorithm. In this regard, the SAOM is commonly described as an agent-based model that allows for statistical inference! In addition to Method of Moments estimation, SAOMs can be estimated by Maximum Likelihood Estimation, Generalized Method of Moments, and Bayesian Estimation (see the RSiena manual for more information on these different techniques [Ripley et al. 2026]).

Core modeling considerations

Specifying an SAOM. The first consideration for fitting stochastic-actor oriented models is de-

termining what endogenous (network-related) and exogenous (actor-related) effects to include in the fitted model. I rely on Block (2015) to facilitate substantive interpretations of the following effects. First, it is commonly assumed that the rate parameter, $\lambda_i(x,z)$, is constant across actors in the network, such that all actors have an equal probability of getting an opportunity to evaluate their outgoing ties. Recent work, like Leal et al. (2026) and Block (2014), have relaxed this assumption and have specifed the rate parameter as a function of actor-level covariates, z. Second, by default, outdegree (density) and reciprocity are included as effects, since they are fundamental social processes for understanding the evolution of social networks. In inferential network models, outdegree is considered the baseline probability to form and maintain outgoing social ties and is commonly described as the model intercept. Reciprocity represents the tendency for actors to reciprocate nominations, that is, to maintain and create ties to actors that have nominated them. Third, one should include effects to capture local and triadic structures. There are many ways to control for these structures in the inferential network models. Here are some possible ways: (1) transitive triplets, which represents the tendency to create and maintain outgoing ties to friends-of-a-friend, (2) three-cycles, which represents the tendency "to be tied to those who perceive ego as a friend-of-friend" (Block, 2015, 168), and (3) transitive reciprocated triplets, that is, the tendency 'to have reciprocated ties to friends-of-friends." (Block, 2015, 168). We usually include a combination of the two effects: (a) transitive triplets + transitive reciprocated triplets and (b) transitive triplets + three-cycles (Leal, 2021).



(1) Transitive triplets **(2) Three-cycles** **(3) Transitive reciprocated triplets**

Fourth, one should capture node-level heterogeneity in the degree-related distributions. This is quite important, since not controlling for node-level heterogeneity in degree distributions can

cause ghost triadic and local structure effects (i.e., rejecting the null when the true value is 0; Type I errors) (Faust, 2007; Juozaitienė and Wit, 2024). As Snijders, Van De Bunt and Steglich (2010) notes, one should include indegree popularity, outdegree activity, and one of the two: indegree activity or outdegree popularity. Per Block (2015), (1) indegree popularity is the tendency for actors to nominate others who are popular (i.e., have a high indegree), (2) outdegree activity is the tendency for actors who have many outgoing ties to nominate more alters, (3) indegree activity is the preference for actors who have high indegrees (i.e., popular actors) to nominate more actors, and lastly, (4), outdegree popularity is the tendency for actors to nominate other actors who have nominated many other actors. Fifth, it is often practical to include node- and dyadic-level effects to control for the influence of exogenous actor-level attributes on network evolution. The SAOM models node-level attributes as ego (outdegree) and alter (indegree) effects. Ego effects account for the influence of *i*'s attribute (e.g., gender) on their outgoing ties, that is, the association between node-level attributes and the outdegree distribution, whereas alter effects test the association between node-level attributes and the indegree distribution, namely, the influence of *i*'s attribute on their incoming nominations. For example, if gender is the focal attribute under analysis (male = 1; female = 0), a *male alter* effect tests if male nodes are more likely to receive ties in comparison to females, while the *male ego* effect tests if male nodes are more likely to send ties in reference to females. Lastly, it is often of importance to include dyadic similarity effects to test for the creation and maintenance of homophilous dyadic ties (e.g., same gender, socio-economic status similarity, same neighborhood). For example, the same gender effect would test if same gender ties are more likely to be created and maintained, in comparison to ties amongst heterophilous dyads.

Model goodness-of-fit. The second consideration of fitting stochastic actor-oriented models is assessing the model goodness-of-fit. Here, we are interested in how well the fitted model captures the underlying data generating process? In other words, does the model recreate the observed network structure well (e.g., indegree distribution)? Following Lospinoso and Snijders (2019), the goodness-of-fit of a fitted SAOM is assessed by generating a set of simulated networks for wave *m* based upon the fitted model and checking the Mahalanobis distance for each computed statistic.

In cases of two waves of data, m is set to 2. With respect to the indegree distribution, we can compare the count of the number of actors with indegree of 3 ($x_{+j} = 3$) in the simulated networks to the number of actors with indegree equal to 3 in the observed network. It is important to note that Mahalanobis distance p-values are computed for each distribution and that smaller p-values indicate worse GOF. In addition, there is a functional difference between a p-value close to 0 and a p-value equal to 0. More specificially, a p-value equal to 0 indicates that the set of observed data points are "far away" from the distribution of simulated statistics, while a p-value close to 0 suggests that the set of observed data points is within the distribution of simulated statistics, but at the extreme of the distribution (Lospinoso and Snijders, 2019).

Time heterogeneity (for more than two waves). The third modeling consideration occurs when we have more than two waves of data collection. An earlier assumption in continuous-time Markov models is *time homogenity* in the process, such that the effects of the network structure is constant across time ($\forall_{t\in\mathbb{R}}\beta_k(t) = \beta_k$). This assumption may not hold in all cases. Let's say we have collected data on best friendship dynamics in a U.S. high school across three waves of data collection (wave 1: 9th grade; wave 2: 10th grade; wave 3: 11th grade). In this case, we have two unobserved periods of network change (i.e., period 1: $9 \rightarrow 10$; period 2: $10 \rightarrow 11$). In general, we would fit one SAOM to both periods and obtained a set of parameters ($\hat{\beta}$) that represent the 'importance' of the specified effects for the evolution of the network. Importantly, in finding the set of parameters, $\hat{\beta}$, the SAOM assumes that the effects are homogenous (the same) across both periods (i.e., the effect of reciprocity on creating and maintaining friendship ties is the same for period 1 and period 2). However, what if gender homophily is 'more' important during period 1 for creating and maintaining friendship ties, in comparison to period 2. In this case, the effects are heterogenous across time. Lospinoso et al. (2011) provide a method for assessing time heterogeneity when there are more than three waves of data collection! How can we fix time heterogeneity? Lospinoso et al. (2011) propose to either split the analyses by period or to include an interaction between the period and the network effect to capture the heterogeneity across periods (Lospinoso et al., 2011). One should prefer the inclusion of interactions, since it preserves more variation by having multiple

periods (and increases the statistical power for the fitted model), as long as there are not too many effects that are heterogenous across periods.

Model misspecification. The fourth modeling consideration is considering model/effect misspecification. In the standard stochastic actor-oriented modeling procedure, effects are estimated as standard evaluation parameters, which assume that the specified effect (e.g., reciprocity, male ego) is the same for maintaining and creating outgoing ties. In other words, without loss of generality, one can think of the evaluation effect for some parameter k, $\beta_{\text{evaluation}}$, as being the average of the maintenance effect, β_{maintain}, that is, the 'importance' of the effect for maintaining outgoing ties, and the creation effect, β_{create}, namely, the 'importance' of the effect for creating outgoing ties. The question remains: does the effect differ for tie creation and tie maintenance? For instance, is gender homophily 'more important' for maintaining the $i \rightarrow j$ tie than creating the $i \rightarrow h$ tie? Block (2014, 97-119) and Leal et al. (2026) show that the standard assumption that the effects are the same for both processes can lead to null and/or disparate findings when the effect differs for tie creation and maintenance! In particular, Leal et al. (2026) show that via simulations, if $\beta_{\text{create}} > 0$ and $\beta_{\text{maintain}} < 0$, then the standard tie evaluation effect will lead to findings of a null effect. However, if β_{create} and β_{maintain} are in the same direction (e.g., both positive), then the estimated standard evaluation effect will reproduce the underlying data generating process. Therefore, it is important to think critically about how the included effect influences the evolution of your empirically observed networks!

Missing data and composition changes. The fifth modeling consideration is related to missing data and composition changes. Like in standard regression analysis, missing data can be problematic for fitting an SAOM. In inferential network models, we have two types of missing data: (a) missing node attribute data and (b) missing tie data. First, missing tie data can be related to item non-response or related to the fact that actors may enter and/or leave the network over time. The latter type of missing tie data relates to composition changes. Currently, the SAOM provides the *structural zeros* and the *method of joiners and leavers* approaches to control for handling composition changes. In relation to missing node attribute data, by standard practice, the SAOM mean

imputes the average values for the purposes of simulation, but does not include these data points for the updating of parameter values. However, more than 10% of missing data for an attribute can be quite problematic (see the RSiena manual). I suggest that you look at the RSiena manual for different procedures for handling missing data! As Duxbury (2023) points out, list-wise deletion should be the last option for handling missing network data. In standard regression analyses, list-wise deletion of n actors who have missing data results in the removal of n data points, but in network models, removing n actors results in the removal of $n(n-1)$ data points. (The removal of $n(n-1)$ data points relates to the dropping of all outgoing and incoming ties for the n removed actors.)

Statistical power. The sixth modeling consideration is concerned with the statistical power of the fitted model Stadtfeld et al. (2020). Let $\pi_k(g,W) = \text{P(Reject } H_0 | \beta_k \neq 0)$ denote the statistical power of the effect k for a fitted stochastic actor-oriented model. Here $\pi_k(g,W)$ is a function of g, the number of actors in the observed dataset, and the number of observational waves W. In a simulation study across 200 replications, Stadtfeld et al. (2020) found that the statistical power for dyadic similarity (homophily) effects is:

- $\pi_k(g,W) = 0.345$ when $g = 20$ and $W = 2$.
- $\pi_k(g,W) = 0.995$ when $g = 120$ and $W = 2$.
- $\pi_k(g,W) = 0.995$ when $g = 60$ and $W = 3$

However, for peer influence effects from co-evolution models (i.e., models that have a behavioral and network dependent), the statistical power for the average peer similarity effect is:

- $\pi_k(g,W) = 0.100$ when $g = 20$ and $W = 2$
- $\pi_k(g,W) = 0.345$ when $g = 120$ and $W = 2$
- $\pi_k(g,W) = 0.345$ when $g = 60$ and $W = 3$

Therefore, it is important that when drawing conclusions from a fitted SAOM, we interpret our findings with respect to the number of actors and waves of data collection. That is, an insignificant

effect (assuming traditional hypothesis testing of a cutoff p-value set to 0.05) may be a result of inadequate statistical power (or also, model misspecication!).

Stochastic simulations. The final modeling consideration is concerned with the stochastic procedure that is used to find the set of Method of Moments parameter estimates. Recall that the SAOM finds the set of parameter estimates with an agent-based simulation approach, which means that the estimated values will vary over estimation runs. As the RSiena manual and Duxbury (2023) points out, one can set the random number seed to replicate their results across simulation runs. It is good practice to check the parameter estimates from your fitted model to those estimates found across different random number seeds (e.g., 37, 3718, 9999).

All that is glitter is not gold

While the SAOM is a powerful inferential network model for studying network and behavior dynamics, recent work has highlighted three potential limitations of the SAOM: (1) the scalability of the model, (2) coordination and strategic settings, and (3) valued and weighted ties (Leifeld and Cranmer, 2019). First, in terms of scalability, as the number of actors in the network, g, increases, the assumptions of the actor-oriented framework become harder to justify. Imagine we have an observational dataset of with 800 students, is it possible for the chosen actor to have knowledge about the *complete distribution of possible ties and actor attributes*? Second, in some research settings, the assumption of only one tie changing at a time may be problematic (see Mayer [1984] for an early critique of this assumption). For example, in coordination and strategic settings, such as political alliance networks and collective action networks, multiple actors may change their outgoing ties simultaneously (Leifeld and Cranmer, 2019). Third, currently, the SAOM does not allow for valued (or weighted) ties to be modeled. As a consequence, we cannot model how endogenous network effects and actor-level attributes impact the strength of ties.

Table 1: Relevant Literature on the Development and Applications of the SAOM

Topic	*Notable References*
RSiena manual:	Ripley et al. (2026) or click here: RSiena Manual
Continuous-time markov models for social network data:	Holland and Leinhardt (1977), Wasserman (1980), Leenders (1995)
Stochastic-actor oriented model for network dynamics:	Snijders (1996), Snijders (2001), Snijders, Van De Bunt and Steglich (2010), Steglich and Snijders (2022), Kalish (2020)
Stochastic-actor oriented model for the co-evolution of behavior and network dynamics:	General overview: Steglich, Snijders and Pearson (2010), Snijders, Van De Bunt and Steglich (2010), Kalish (2020) Diffusion of Innovations: Greenan (2015) Continuous behaviors: Niezink, Snijders and van Duijn (2019)
Handling missing data and composition change:	Huisman and Snijders (2003), Huisman and Steglich (2008), Zandberg and Huisman (2019), de la Haye et al. (2017), Hipp et al. (2015)
Assessing time heterogeneity:	Lospinoso et al. (2011)
Model misspecification:	Block (2014, 97-119), Leal et al. (2026)
Checking model goodness of fit:	Lospinoso and Snijders (2019)
Notes on statistical power:	Stadtfeld et al. (2020)
Methods for handling multiple networks:	Snijders and Baerveldt (2003), An (2015), Koskinen and Snijders (2007)
Two-mode networks:	Fujimoto, Snijders and Valente (2018)
Network visualizations (helpful for understanding microsteps):	adams and Schaefer (2018)
Computational Experiments with SAOMs:	Steglich and Snijders (2022), McMillan and Schaefer (2021), Duxbury and Haynie (2020), An (2015)
Comparison and Critiques of the SAOM †:	Ragan et al. (2022), Leifeld and Cranmer (2019) Block et al. (2022), Leifeld and Cranmer (2022), Wasserman and Brandes (2022)

Notes: The bibliography for the references shown in this table can be found in the references at the end of this document. †: I'm grateful to Eunsung Yoon for sharing these articles in an earlier discussion.

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